4.7 Active Directory

What is Active Directory?

Directory service for domain networks

Used for centralized domain management and directory-based identity services

Runs on domain controllers

New Key Ideas:

- Domain users and groups
- Group policy management
- Better organization via schema
- More default shares
  - SYSVOL
  - NETLOGON
- Forest, domain tree structures
- Kerberos
- Trusts
- A lot more stuff, some unrelated to security

FSMO types

- ADDS
  - Domain Services - authentication services
- ADLDS
  - Lightweight Directory Services - stand alone, multiple instances, but less functionality
- Other ones that you'll never see again (hopefully):
  - ADFS
  - ADCS
  - ADRMS

Directory Structure

Everything is an **object**

Objects are either *resources* like printers or *security principals* like users or groups, which all have **attributes**

Schema defines object content and structure

Containers and OUs are the equivalent of a folder

Forest, domain tree structure

Forest

- set of domain tree(s) that share a common schema, configuration, and global catalog

Domain tree

- set of domain(s) that share a common schema and configuration

Ex: forest may include abc.com and xyz.com domain trees

Domain tree may include eu.abc.com and us.abc.com

New Users!

Domain Users

- You can log in as any domain user on any computer in the domain

Domain Admins

- Domain Users but Admins woahhh

There are many admins

9 built-in groups with the word "Admin" in them

Administrators

- Have administrative access to DC

Domain Admins

- Have administrative access to all in domain

Enterprise Admins

- Have administrative access to all in forest

Schema Admins

- Have admin privileges to change the whole schema (they can change the framework)

… and a lot more

Check these groups for unauthorized

AD Users

AD Users have a lot more properties. We will explore these in the lab, along with AD computers.

Local Users no more!

Domain Controllers do not use local users, but instead domain users.

NO lusrmgr.msc

Instead, use dsa.msc

> Same as Server Manager > Tools > Active Directory Users and Computers

Group Policy Management

Active Directory Domain Controllers can assign different group policies to different computers with gpmc.msc

- Ex: enforce so that server machines are more restricted than workstations

Load/configure the group policies and assign them to the OUs (containers of objects) you want

We will also explore this in the lab

Group Policy Preferences

You can change the order in which group policies are applied to a computer or OU

Not very important, but good to know